

■ Plum OPD — AI-Powered Adjudication

Internship Project Submission • AI Automation Engineer

Project Overview

Plum OPD is an intelligent, full-stack web application that completely automates the approval and rejection of outpatient medical insurance claims. Designed to eliminate manual processing delays, the tool accepts bill or prescription scans (PDF/images), runs optical character recognition (OCR), and uses Google Gemini AI to parse unstructured text into clean medical metadata. A policy-driven rule engine then evaluates the claim against 8 automated rules in priority order. If the claim is flagged with anomalies or potential duplicates, it is routed to a custom Admin Manual Adjudication queue where administrators can perform overrides (approve with custom amounts or reject with notes) in real-time.

Technical Stack

Frontend: React 19, Vite 8, React Router 7, Axios, Monochromatic Custom CSS Theme

Backend: Node.js, Express 5, Multer File Uploader

AI & OCR Engine: Google Gemini 2.5 Flash API (structured JSON output), Tesseract.js 7

Database: MongoDB Atlas (Mongoose 9)

Deployments: Vercel (Frontend), Railway (Backend API)

Project Delivery Links

GitHub Repository	github.com/AnveshCheela/plum-opd-project
Live Web Application	plum-opd-project.vercel.app/
Source Code (Google Drive)	Drive Folder (Zip & Scans)
Video Walkthrough (YouTube)	youtu.be/Pe6P37RdJ3M